Structural Estimation, Sufficient Statistics, and Calibration

Week 1

Student-led Structural Estimation Workshop

HKUST(GZ)

Today's learning

- **Structural estimation** recovers model parameters and simulates the policy.
- **Sufficient statistics** identify only the empirical objects needed for a welfare formula.
- **Calibration** selects reasonable parameter values and treats the model as a quantitative experiment.

The methods differ in how much structure they impose and which empirical objects they require.

Structural approach

What does the model add?

A model is a probability distribution over outcomes

- Sargent's definition: a model is a probability distribution over a sequence of outcomes, possibly indexed by parameters.
- Economic models add variables and restrictions that represent choices, technology, information, and equilibrium.
- The usual objective is a counterfactual: what happens to Y when policy or environment X changes?
- **Applied micro:** Counterfactuals often use quasi-experimental variation plus statistical assumptions.
- **Quantitative models:** Counterfactuals combine structural assumptions with parameter values.

The feedback effects

- Structural analysis becomes valuable when outcomes and explanatory variables respond to each other.
- Supply and demand or general equilibrium are natural examples.
- Nonlinearity often makes structural models more useful for counterfactuals, although reduced-form models can also support counterfactual analysis.

Feedback is one reason to use structure, not a complete definition of structural work.

Structural estimation compresses data into parameters

- The model maps a few structural parameters into many behaviors.
- Estimation reverses that map: observed behavior disciplines the parameters.
- Counterfactuals then use the estimated parameter vector in a new environment.

Table: Structural Parameter estimates

Parameter	Estimate
θ_1	$X.XX$
	$(0.XX)$
θ_2	$X.XX$
	$(0.XX)$
$\vdots$	$\vdots$

Standard errors are inside $()$.

Structural Model Structure

Following Matzkin (2007), a general formulation of a structural model is

$$\mathcal{M}(Y, X, \varepsilon; F, G) = 0$$

- Y denotes an observed vector of endogenous variables (choices and outcomes).
- (X, ε) are vectors of observed and unobserved variables.
- F is a vector of structural functions.
- G is a vector of distributions for random variables.
- $\mathcal{M} = 0$: a vector of structural relations between the primitive objects.

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The goal of structural estimation is not to estimate a single reduced-form coefficient, but to recover the primitives that generate behavior and connect them to data through optimization, constraints, and stochastic processes.

A labor supply example

The household's problem:

$$\begin{aligned}\max_{c_i, l_i, h_i} \quad & U(c_i, l_i, X_i, \varepsilon_i^u) \\ \text{s.t.} \quad & c_i = w_i h_i + I_i, \\ & T_i = h_i + l_i, \\ & w_i = \exp(\alpha_X^w X_i + \varepsilon_i^w).\end{aligned}$$

- The individual trades consumption against leisure by choosing hours h_i.
- X_i, non-labor income I_i, and available time T_i are observed exogenous variables.
- ε_i^u captures unobserved distaste for work and ε_i^w captures unobserved productivity.
- The model allows ε_i^u and ε_i^w to be correlated.

The constraints imply a labor supply function

Substituting consumption and leisure into utility reduces the problem to one choice:

$$\max_{h_i} \; U(w_i h_i + I_i, T_i - h_i, X_i, \varepsilon_i^u) \qquad \text{s.t. } w_i = \exp(\alpha_X^w X_i + \varepsilon_i^w).$$

Let the solution be

$$h_i^* = h(w_i, I_i, X_i, \varepsilon_i^u).$$

- The structural primitives are the utility function, wage equation, and joint distribution of unobservables.
- The labor supply function is the model-implied relationship between hours and exogenous variables.
- Consumption and leisure follow from the budget and time constraints once h_i^* is known.

The research question determines how much structure is needed

If the target is the effect of an exogenous wage change on hours, the object of interest is

$$\frac{\partial h(w, I, X, \varepsilon^u)}{\partial w}.$$

Credible wage variation may identify this response without recovering the entire utility function.

A new nonlinear income tax changes the budget set:

$$c_i = w_i h_i + I_i - \tau(w_i, h_i).$$

Brackets, marginal rates, and notches can generate responses outside the variation observed in the original data.

The model should contain only the structural features needed for the proposed counterfactual.

Formulation begins with agents and interactions

Agents

- A single-agent model treats each individual as an isolated decision maker.
- A household labor-supply model may require separate utility functions and wages for both spouses.
- Joint decisions require a bargaining rule, Pareto-efficiency restriction, or non-cooperative game.

Equilibrium

- Partial equilibrium takes wages as given and models labor supply in detail.
- An equilibrium model adds labor demand and imposes

$$H(w) = \sum_i h_i(w_i) = D(w).$$

- Equilibrium wages may offset part of the hours response to a large tax or transfer reform.

Dynamics require a horizon, expectations, and a time unit

Planning horizon

- Are agents myopic or forward-looking? Is the horizon finite or infinite?
- Forward-looking behavior introduces a discount factor and continuation values.

Uncertainty

- Future heterogeneity may be known in advance or arrive as an unanticipated shock.
- The distinction changes expectations, optimal choices, and the likelihood.

Time unit

- A period may be a year, month, week, or decision opportunity.
- The time unit determines how discounting, job offers, depreciation, and transitions are interpreted.

Choice sets and state variables determine the solution problem

Choice set

- Hours may be continuous, or participation may be represented by $d_i \in \{0, 1\}$.
- In a binary work model, an individual works when the utility of work exceeds the utility of non-work.
- Discretization can simplify computation but may require stronger identifying assumptions.

State space

- $\Omega_{i,t}$ contains everything known and relevant when the agent chooses.
- Age evolves exogenously. Experience evolves through past work decisions.
- Dynamic models also require a transition law $\Pr(\Omega_{i,t+1} \mid \Omega_{i,t}, d_{i,t})$.

The objective and information set must be explicit

Objective function

- Expected discounted utility with exponential discounting is common, but it is a modeling choice.
- Alternatives include non-expected utility or time-inconsistent preferences.

Information

- Separate what the decision maker observes from what the econometrician observes.
- In the labor-supply example, the worker observes tastes and productivity that the econometrician may not observe.

Observed heterogeneity

- Primitive objects may vary with education, age, or other observed characteristics.
- Holding some primitives fixed across groups can create exclusion restrictions used for identification.

Unobserved heterogeneity and uncertainty are different

A dynamic model may decompose the taste shock as

$$\varepsilon^{u}_{i,t} = \mu^{u}_{i} + \nu_{i,t}.$$

- μ^{u}_{i} is persistent heterogeneity known to the individual.
- $\nu_{i,t}$ is a new shock that was not known before period t.
- A finite-mixture model may represent μ^{u}_{i} through latent types and type probabilities.
- Confusing permanent heterogeneity with shocks changes both behavior and parameter identification.

The distribution of types and shocks becomes part of the structural model and must be disciplined by data or external restrictions.

Functional forms and distributions complete the model

A parametric labor-supply application may specify

$$U(c_i, l_i, X_i, \varepsilon_i^u) = \frac{c_i^{1+\eta}}{1+\eta} - \exp(\psi X_i + \varepsilon_i^u)\frac{(T_i - l_i)^{1+\gamma}}{1+\gamma},$$

with $\eta < 0$ and $\gamma > 0$.
It must also specify the joint distribution of unobservables, for example

$$\begin{pmatrix} \varepsilon_i^u \\ \varepsilon_i^w \end{pmatrix} \sim N\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \sigma_u^2 & \rho\sigma_u\sigma_w \\ \rho\sigma_u\sigma_w & \sigma_w^2 \end{pmatrix} \right).$$

When $\rho \neq 0$, wages are endogenous in the labor-supply equation because $E[\varepsilon_i^u \mid w_i] \neq 0$.

Investigating Identification

Identification: Under the model $\mathcal{M}(\cdot) = 0$, whether there exists a unique set of structures (F, G) that can generate the observed $\Pr(Y, X)$. If two distinct structures (F, G) and (F', G') can both produce the same data distribution, then the relevant primitives are not identified.

Formal identification proof is difficult for complex dynamic models, and applied papers typically provide several types of supporting evidence.

- Present an identification map: which variations, moments, or restrictions identify which parameters.
- Check local identification: whether the likelihood is flat and whether the moment Jacobian is near-singular.
- Conduct sensitivity analysis: whether moments and counterfactuals remain stable when key parameters or assumptions are altered.
- Use Monte Carlo recovery to verify whether the estimation procedure can recover the true values from simulated data.

Estimating structural models methods

Classic approasch, simulation-based estimation, unobserved heterogenity, CCP estimation (Hotz and Miller, 1993) and measurement error.

From the perspective of object function:

- Match the full probability or density,
- Match the moments,
- Match the auxiliary statistics from simulated data,
- Leverage the conditional choice probabilities of dynamic discrete choice.

We will talk about them in detail next several weeks...

Sufficient statistics

How much structure does the welfare question require?

Two paradigms leave a gap

Structural approach

- Estimate primitives of a model.
- Simulate the welfare effects of policies.
- Concern: the full primitive structure may require strong assumptions.

Reduced-form approach

- Use transparent exogenous variation to estimate treatment effects.
- Concern: treatment effects may change with the policy regime.
- A treatment effect alone may not deliver a welfare measure.

The sufficient-statistics bridge

$$\text{Structural primitives} \longrightarrow \text{Sufficient statistics} \longrightarrow \text{Welfare change}$$

$$\begin{aligned} \text{Preferences and constraints} \quad & \omega = (\omega_1, \ldots, \omega_N), \\ \text{Behavioral responses} \quad & \beta = f(\omega, t), \\ \text{Policy target} \quad & \frac{d\mathcal{W}}{dt} = g(\beta, t). \end{aligned}$$

The response β can be estimated with program-evaluation methods.

Different primitive vectors can yield the same sufficient statistics and therefore the same local welfare result.

Harberger's commodity-tax setup

The consumer chooses produced goods $x = (x_1, \ldots, x_J)$ and a numeraire y:

$$\max_{x,y} u(x) + y$$
$$\text{s.t. } px + tx_1 + y = Z.$$

Competitive firms solve

$$\max_{x} \; px - c(x).$$

Social welfare equals consumer utility, profits, and tax revenue (eliminating price p):

$$\mathcal{W}(t) = \max_{x}\{u(x) + Z - tx_1 - c(x)\} + tx_1.$$

The envelope theorem removes most primitives

Private choices maximize the expression inside braces:

$$\mathcal{W}(t) = \underbrace{\max_{x}\{u(x) + Z - tx_1 - c(x)\}}_{\text{private surplus}} + tx_1 .$$

The envelope theorem gives

$$\frac{d\mathcal{W}}{dt} = -x_1 + x_1 + t\frac{dx_1}{dt} = \boxed{t\frac{dx_1}{dt}}.$$

- The existing tax wedge t is observed.
- The demand response dx_1 / dt can be estimated using credible policy variation.
- The analyst need not recover the entire demand and supply system for this local welfare derivative.

Chetty's six-step rubric

1. Specify a general model structure.
2. Represent $d\mathcal{W}/dt$ using multipliers.
3. Map multipliers to marginal utilities using first-order conditions.
4. Recover marginal utilities from observed choices.
5. Estimate the welfare formula empirically.
6. Use a structural model to evaluate approximations and extrapolation.

The approach reduces the number of primitives to estimate. It does not make welfare analysis theory-free.

The sufficient statistic is target-specific

Discussion

The elasticity dx_1 / dt is sufficient for which object, under which market assumptions, and for how large a policy change?

The simple formula needs more structure when the application includes:

- externalities or imperfect competition;
- distributional welfare weights;
- a large reform that moves behavioral responses along the policy path;
- general-equilibrium price changes.

Sufficiency is a property of a statistic relative to a target and a model class.

Advantages and limits of sufficient statistics

Advantages

- Fewer parameters need identification.
- Reduced-form designs can identify the relevant behavioral responses.
- Rich heterogeneity may collapse into an aggregate response.

Limits

- Results are usually local around observed policies.
- The formula can hide competing mechanisms.
- Global or optimal policy analysis often needs a structural model.

The two approaches work well together

- Use sufficient-statistics formulas to benchmark a model's local welfare prediction.
- Ask whether one structural parameter vector matches several independently estimated sufficient statistics.
- Calibrate or estimate the model using policy-relevant moments.
- Use the structural model for policy changes outside the local empirical support.

Credible local response $+$ disciplined extrapolation

Calibration

How should data discipline a quantitative model?

Why researchers calibrate

Canova (2007) describes several motivations:

- The data may not identify every parameter statistically.
- Large misspecification can make formal estimates economically unreasonable.
- Researchers may combine economic restrictions with empirical evidence rather than treat the entire model as the true data-generating process.

Calibration is therefore a collection of procedures for answering quantitative questions with models that researchers know are approximations.

The calibration procedure

The essence of calibration by Kydland and Prescott (1991, 1996)

1. Choose the economic question to be addressed.
2. Select a model design which bears some relevance to the question asked.
3. Choose functional forms and solve endogenous variables as functions of parameters and shocks.
4. Select parameters and convenient specifications for the exogenous processes and simulate paths for the endogenous variables.
5. Evaluate the quality of the model by comparing its outcomes to a set of ”stylized facts” of the actual data.
6. Answer the question and characterize uncertainty around the answer.

Calibration and estimation can look mathematically similar

A common calibration chooses θ to minimize moment discrepancies:

$$\widehat{\theta} = \arg\min_{\theta} \sum_{i=1}^{n} W_i \left[\log \mu_i(\theta; \psi) - \log \widehat{\mu}_i\right]^2 ,$$

where ψ is a vector of “deep parameters” chosen in the previous step and W_i some arbitrary weight.

- With a statistical weighting rule, the criterion can coincide with SMM or minimum distance.
- Calibration differs mainly in how evidence enters and how uncertainty is reported.

Calibration in practice

1. Determine parameters directly from data outside the model.
2. Borrow relatively stable parameters from other studies.
3. Choose remaining parameters to match model moments with data moments.
4. Validate the model using moments excluded from parameter selection.

Informative moments move with the right parameters

- Comparative statics should show which parameter changes which moment.
- Level moments rarely provide strong information about elasticities or curvature.
- Causally estimated responses and impulse responses can discipline a specific mechanism.
- A useful moment responds strongly to the parameter of interest and differently from moments tied to competing mechanisms.

Matching as many moments as parameters is necessary for exact matching, not sufficient for credible identification.

Targeted moments and validation moments play different roles

Targeted moments

- determine free parameters;
- may match exactly by construction;
- reveal whether the calibration problem can be solved.

Non-target moments

- test related implications not used in selection;
- reveal missing mechanisms;
- can include replication of an empirical treatment effect.

Parameter uncertainty can be smaller than model uncertainty

Jon Steinsson: “Calibration is just moment matching without standard errors.”

The point is not that standard errors are irrelevant. The larger concern may be that the model omits an important mechanism.

- Parameter uncertainty asks how much the chosen moments pin down θ.
- Model misspecification asks whether the model class can answer the policy question.
- Sensitivity should vary both parameter values and consequential model assumptions.

The approaches answer different questions

Approach	Main output	Source of credibility	Main limitation
Reduced form	Treatment effect in an observed setting	Research design and estimand	Limited transport to new regimes
Sufficient statistics	Local welfare effect	Causal response plus economic restrictions	Target and model-class specific
Structural estimation	Primitives and counterfactual distributions	Identification, invariance, equilibrium, validation	Stronger modeling and computation
Calibration	Quantitative model experiment	Transparent parameter evidence, non-target fit, sensitivity	Discretion and model misspecification

The approaches often complement one another rather than compete.

Resources

- Henrik Kleven's teaching materials: `https://www.henrikkleven.com/lectures`
- Tomas Martinez's course about calibration in macro: `https://tomasrm.github.io/teaching/quantmacro/data_macro.pdf`

References

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